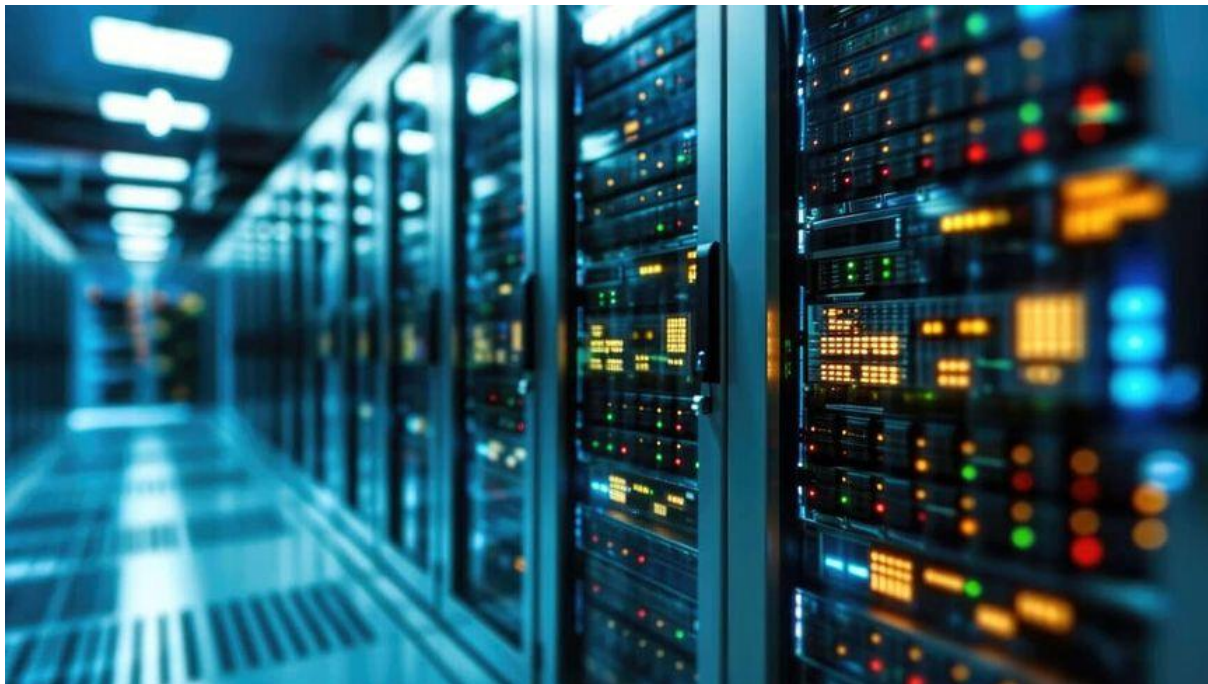


Introduction to Datacenters-17

A **Data Center** is a **physical facility** that contains networked computers, storage systems, and other equipment used to organize, process, store, and distribute large volumes of data.

- It is a **specialized facility** that organizations use to house their critical applications, services, and IT infrastructure.
- Ensures high availability of IT services.
- Provides secure, reliable, and scalable infrastructure for businesses, cloud computing, and online services.
- Key components include servers, storage solutions, networking equipment (like switches, routers, and firewalls), and essential infrastructure for power supply, cooling, and physical security.
- Powers **business continuity** with disaster recovery.



Servers

- High-performance computers that run applications and process data.
- Can be physical (bare metal) or virtual (VMs, containers).
- Examples: Web servers, database servers, mail servers.

Networking Equipment

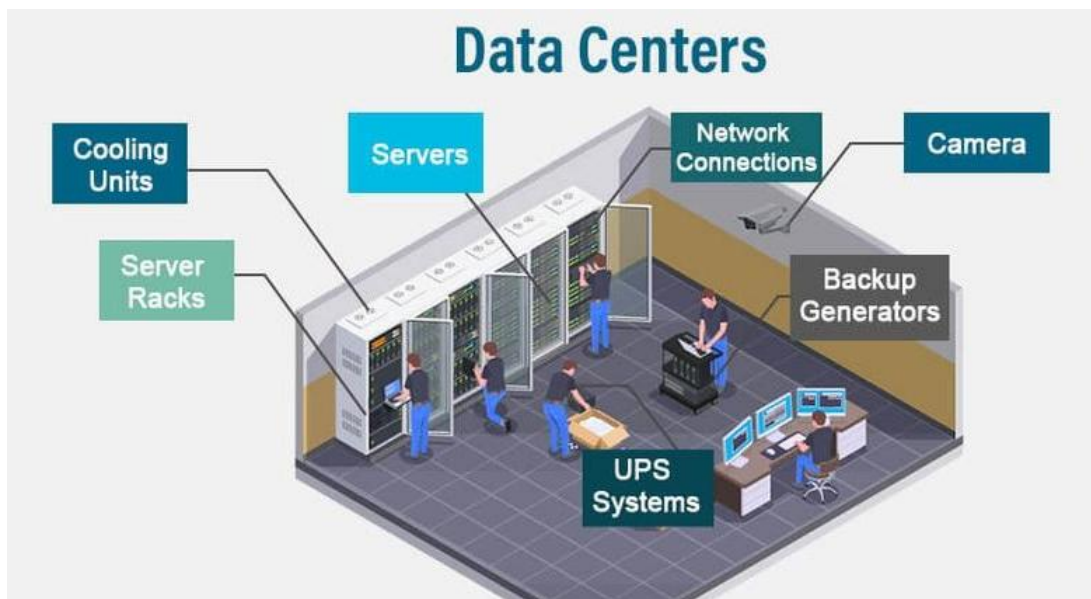
Connects servers, storage, and users Includes:

- **Switches** – connect devices within the data center.

- **Routers** – connect the data center to external networks (internet, WAN).
- **Firewalls** – protect against unauthorized access.
- Ensuring smooth communication and security.

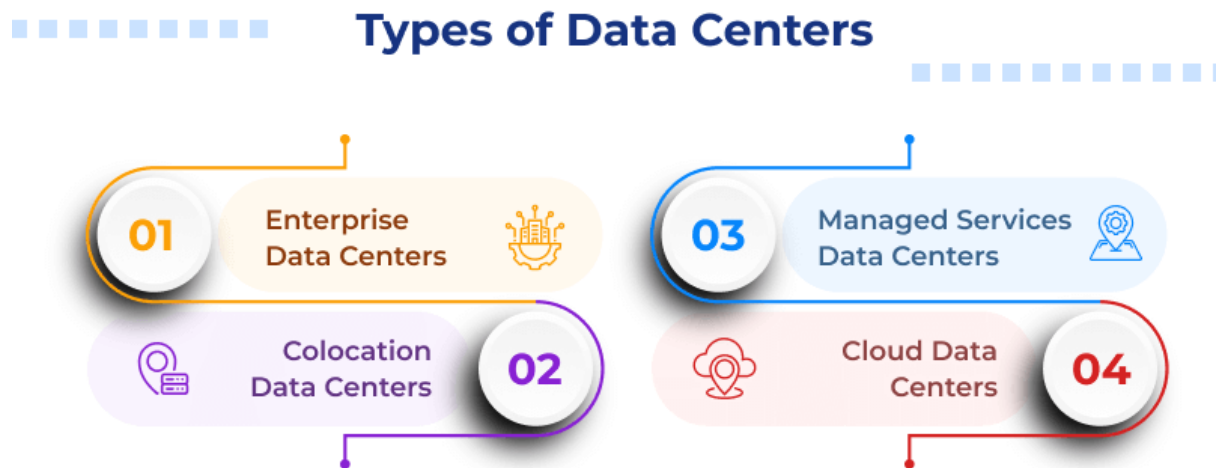
Storage Systems

- Devices and infrastructure for storing large volumes of data.
- Types:
 - DAS (Direct-Attached Storage)- Storage directly connected to a server which are Simple and low-cost.
 - SAN (Storage Area Network)- High-speed **dedicated network** that connects servers to storage devices which Provides block-level storage (like a virtual hard disk).-
 - NAS (Network-Attached Storage)-Storage device connected to a standard **IP network (Ethernet)** which provides file-level access (like shared folders) Easy to expand and manage.
- Ensures fast access, backups, and data recovery.



Types of Data Centers

- **Enterprise Data Center:** Owned/operated by a company.
- **Cloud Data Center:** Managed by cloud providers (AWS, Azure, GCP).
- **On-Premises Data Center:** managed by an organization within its own physical location.



CIA

Confidentiality, Integrity, and Availability triad. It's a core security model used to protect data and systems.



- **Confidentiality**

Ensures only authorized people or systems can access data. In data centers, this means access controls, encryption, and authentication to keep sensitive data private.

- **Integrity**

Ensures data is accurate, consistent, and not tampered with. Techniques include hashing, version control, and secure transfer protocols to detect and prevent unauthorized changes.

- **Availability**

Ensures data and services are accessible whenever needed. In data centers, this involves redundancy, backups, failover systems, and disaster recovery plans to minimize downtime.

Together, the CIA triad forms the foundation of **data center security policies** and is used to guide design, operations